

Classification of Challenges in Achieving BIM-Based Safety-Requirement Checking in Vertical Construction Projects

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Abstract: Due to the one-off nature of projects, which necessitates setting up of unique production sites, the construction industry requires stringent requirements for safety of the workforce and the public around job sites. Although there are various federal (e.g., OSHA) and local safety regulations (e.g., NYC Department of Buildings codes) to keep construction sites safe and incident free, around 20% of work fatalities in the private industry have been from the construction industry in the past years. Within the current practice, the requirements listed in these regulations are kept within document-based safety plans. The challenge is the large number of safety requirements to be checked with more than 1,000 sections and multiple narratives under each. Given this, the manual approach fails in tracking whether these requirements are properly considered in safety plans, hence resulting in omissions and accidents/incidents due to inadequate implementations of safety requirements. Although automated safety compliance-checking coupled with building information modeling (BIM) has been an opportunity to bring efficiency to the process, there are several challenges that stand in the way of the ideal practice. This research provides the results of rigorous research conducted with the guidance of practitioners in the construction safety and insurance domain, and analysis and implementation of OSHA Part 1926 narratives using model-based checking tools. The results have been consolidated and presented as a taxonomy of challenges along with potential solutions to these challenges. Researchers and practitioners (software vendors, construction companies, regulators) can build on the outcomes to generate solutions to address these challenges. DOI: [10.1061/JCEMD4.COENG-13119](https://doi.org/10.1061/JCEMD4.COENG-13119). © 2023 American Society of Civil Engineers.

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Introduction

The dynamic nature of construction job sites exposes workers and the public to safety issues and makes the industry one of the most injury-prone industries. Accidents and near misses are observed at job sites despite federal and local regulations. The construction industry accounts for one in five work fatalities in the private industry (BLS 2017), with more than 26,000 construction worker casualties in the US, 40% of which involved falls from height, and 30% of the falls were caused by inadequate, removed, or inappropriate use of fall protection equipment. Despite existing efforts for preventing safety injuries and near-misses, there is still a need for rethinking the traditional practices on construction safety.

Within the current practice, there are various federal [e.g., Occupational Safety and Health Administration (OSHA)] and local safety regulations [e.g., NYC Department of Buildings (DOB codes)] available to ensure the safety of workers and the public at/around construction sites (OSHA 1926; NYC DoB 2014). These regulations provide requirements to be implemented at job sites for maintaining safe working environments. A major problem in the industry is the manual compliance checking of these regulations

in construction safety plans. There is a substantial number of safety requirements to be maintained within these regulations. For example, only Part 1926 of OSHA has 29 subparts (i.e., major category of safety requirements such as “Fall protection systems”) and around 1,400 sections (i.e., individual subdivisions listed under a subpart such as “Scaffolds”), making the manual compliance checking a tedious and error-prone task resulting in omissions, incidents, and accidents. This practice also results in missed opportunities for proactive checking of the required measurements for upcoming construction activities.

Technological improvements, such as the availability of building information models (BIMs) or enhanced scope in industry foundation class (IFC) representation, already provide opportunities and solutions to perform rule-based checking of safety requirements (e.g., Solibri 2021; Zhang et al. 2013; Kim et al. 2016). These tools and developed solutions rely on a prepared BIM to include (or find ways around to depict) the necessary objects and/or their attributes required for model checking. However, a barrier between the construction industry and wide adoption of rule/model-based safety checking is the lack of a clear depiction of challenges in ruleset representations and model preparation for proactive safety compliance checking.

To facilitate adoption of rule/model-based checking of safety requirements, this research aims to identify challenges that hinder model-based safety compliance checking and provide a taxonomy of challenges for practitioners and researchers to consider when preparing models and rulesets for effective and automated safety compliance checking. These requirements have been identified by examining all sections of Part 1926 of OSHA with the consultancy of experts from insurance and construction companies, converting them to rulesets and documenting the challenges in this process

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using various 3D models as an input. Hence, the main contributions of this work are (1) a developed classification of challenges that implementers should be aware of for model-based safety requirements checking in vertical construction projects; and (2) potential solutions to eliminate these challenges so that safety requirements required for the next phase of an ongoing construction can be proactively checked using 3D semantic models.

The current practice of safety planning and checking, and existing body of knowledge in this area, are discussed in the following section. Then, the research methodology used for identifying the challenges in implementing rule/model-based safety checking is described. Next, the research findings in a list of categorized challenges observed in rule/model-based safety compliance checking are presented. A summary of the contributions of this work and discussions on future work are presented in the conclusion section.

Research Background

Current Practice of Safety Planning

Safety planning plays a crucial role in the construction life cycle to avoid severe consequences such as delay, injury, and even loss of lives. However, construction sites are usually addressed as being “underresourced” and “underplanned” in terms of safety planning (e.g., Zhang and Hu 2011; Egan 1998). Personal protective equipment (PPE) is usually used as the last line to protect the workers from an unavoidable hazard (Irizarry et al. 2005). Regulations indicate that other than educating and training workers and providing the PPE, companies need to have more stringent safety and health implementations at sites depending on the nature of construction activities. These regulations are available at federal (e.g., OSHA Part 1926) and local levels (e.g., NYC DOB Chapter 33) for companies to comply with at sites.

Although the federal and local level regulations provide an extensive set of narratives for safety compliance at sites, studies clearly show that the current practice of construction safety planning is document-based and checked through ineffective manual observations at sites (Zhang et al. 2013; Ku and Mills 2010). Safety plans are generated using 2D drawings manually, and then checked by inspectors if they are properly and proactively implemented at sites. This document and experience-based safety compliance checking is dependent on subjective interpretations and thus error-prone. Another challenge is the voluminous nature of safety regulations, which makes it harder to track them for compliance. Furthermore, construction site safety is mainly the responsibility of contractors; however, lack of proper thought process given during the design phase on the safety needs has been an ongoing problem to date (Hinze and Wiegand 1992; Zhang et al. 2013; Behm 2005).

BIMs for Construction Safety

Virtual design and construction (VDC) and computational approaches that reason with BIMs are widely used in the construction industry (e.g., egress analysis, clash detection, 4D, accessibility analysis) while the construction site safety still undergoes the manual and people-centric approaches. With the wider adoption of the technology, the use of BIMs has gained the grounds by allowing information exchange through computer-interpretable formats (Eastman 2009; Staub-French and Khanzode 2007). Gradually, BIM provided the opportunity of bringing automation into safety compliance.

Previous studies to address safety concerns through BIMs are various and can be grouped into studies that leverage models for (1) visualization and interactive learning environments for safety

training (e.g., Li et al. 2018; Vries et al. 2004); (2) hazard identification (e.g., Benjaoran and Bhokha 2010; Zhang et al. 2015); (3) proactive site safety and layout planning (e.g., Sulankivi et al. 2009); and (4) safety compliance checking (e.g., Zhang et al. 2013).

Various research approaches that leverage models and model-checking tools have been proposed in the literature for safety compliance checking. For example, Zhang et al. (2013) presents an automated rule-based checking system by leveraging data stored in BIMs and how future safety planning can be integrated with the generation of work breakdown structures and project schedules. However, these studies rely on models that contain the objects (e.g., scaffolding), attributes (e.g., scaffolding top platform width), and object decompositions (e.g., scaffolding composed of guardrails, toe boards, platforms) that are typically mentioned in safety narratives, and assume that models that contain such data are readily available to contractors. Despite the potential provided by the BIMs in construction safety checking and management, models that contain safety related objects, attributes, and object decompositions are not readily available, and there is still a need to characterize possible challenges in converting initial design models into models that are usable for safety compliance checking. Hence, our work complements the previous research studies that leverage model data for safety compliance checking but significantly differs from them by presenting a missing-but-needed classification of challenges that prevent full deployment of model-based safety compliance checking. This research also provides potential solutions to practitioners and researchers to overcome challenges and prepare models for proactive compliance checking.

Research Method

The objectives of this research were to identify and classify the challenges inherent in the process of rule-based checking of safety requirements, and provide potential solutions for these challenges that practitioners and software vendors can consider during implementation. The challenges have been investigated in the process with respect to the interpretation of narratives in regulations, representation of rulesets, and preparation of models for automated checking of them using models. Fig. 1 shows the overview of the research method. The regulations analyzed in this study included OSHA Part 1926- safety and health regulations for construction. This investigation included (1) screening sections in OSHA Part 1926 to identify whether sections are possible to check with information that could be stored in BIMs; (2) for the sections that are highlighted, analyzing each narrative and labeling them for their potential to be converted to computer interpretable rulesets; (3) converting marked narratives to computer-interpretable rulesets, and recording information needed in BIMs for these ruleset instances; and (4) conducting experiments with model-checking tools to link identified safety requirements to testbed BIMs and recording challenges observed while preparing models for rule checking.

While defining the rulesets, preparing models for rule/model-based compliance checking, and linking models with the rulesets, all the challenges faced that hindered effective linkage of safety requirements to BIMs were documented throughout the process. While some of the narratives could not be represented as computer interpretable rules to check, if there is a model element that a safety narrative ties to, we marked them as safety notifications to be displayed regarding those elements. For example, for the ‘form deck guard rails’ section, one narrative indicates, “Each employee on the face of formwork or reinforcing steel shall be protected from falling 6 feet (1.8 m) or more to lower levels by personal fall arrest systems, safety net systems, or positioning device systems.”

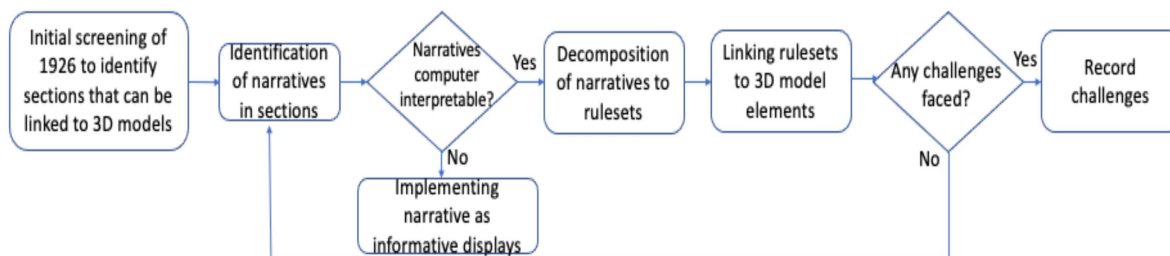
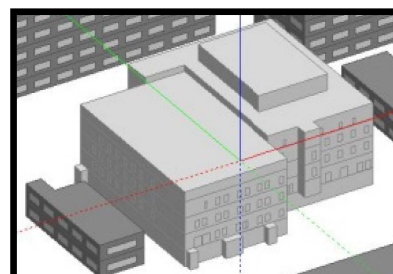


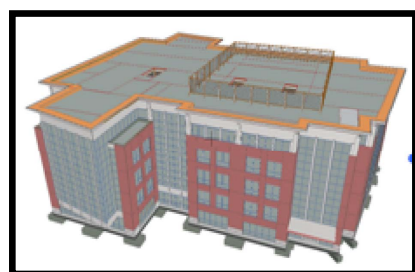
Fig. 1. Overview of the workflow utilized in the study.



~200,000 SF school/administrative building
Reinforced concrete core; Concrete shell



~68,000 office space
Reinforced concrete core; Masonry shell



~300,000 SF commercial space
Steel frame core; Curtain walls and brick masonry shell; Roof with parapets and openings



~500,000 SF office/classroom space
Mixed core (steel and reinforced concrete)
Precast slabs; Stone exterior enclosure

Fig. 2. Overview of models used in the implementation of rulesets.

This narrative will be displayed as informative whenever guard rails are checked for safety implementations, since employee personal protection systems cannot be meaningfully represented in models. Although a significant factor needed in this study to generalize the study results was the breadth of safety requirements analyzed as compared to the variations in 3D models used during testing, we experimented with different 3D models, where variations in the height of buildings, core, and shell systems were observed. In each testbed, at least one of the architectural, structural, mechanical, electrical, and plumbing systems was available to the research team. An overview of the 3D models used in the study is provided in Fig. 2. Documented challenges were then classified based on the extent of modifications needed in BIMs.

Analysis of Safety Regulations

This task included two major subtasks. Subtask 1 included initial screening of the major Federal safety regulation, which is the Occupational Safety and Health Association (OSHA) “Safety and

Health Regulations for Construction” (Part 1926). With Subtask 1, the aim was to generate an initial list of sections within part 1926 that can be linked to BIMs and be automatically checked against a given model. Subtask 2 included detailed analysis of the identified narratives in Subtask 1 to convert those narratives into computer interpretable rulesets.

Initial Screening of OSHA Safety and Health Regulations for Construction (Part 1926)

The purpose of this subtask was to screen through all the safety requirements listed in OSHA Part 1926 to identify safety requirements that can be linked to BIMs. Requirements are listed as narratives, which require laborious work to extract the safety requirements as safety rulesets. OSHA 1926 contains 29 subparts (A through Z and AA through CC). Within these subparts, OSHA 1926 contains around 1,442 sections (with some sections being empty and reserved for future expansion). This effort resulted in a list of sections that directly relate to physical components that are in 3D models (e.g., slabs, openings, exterior enclosure) or

Table 1. List of applicable subparts and example OSHA sections that were analyzed in detail

Applicable OSHA subparts	#	Example sections and narratives identified
C. General safety and health provisions	2/16	1926.34. Means of egress; 34.c. "Means of egress shall be continually maintained free of all obstructions or impediments . . ."
E. Personal protective equipment	3/11 2R ^a	1926.104. Safety belts, lifelines, and lanyards; 105.a. "Safety nets shall be provided when workplaces are more than 25 feet above the ground or water surface . . ."
F. Fire protection	5/10	1926.152. Flammable liquids; 152(d): "At least one portable fire extinguisher, having a rating of not less than 20-B units, shall be located outside of, but not more than 10 feet from, the door opening . . . more than 60 gallons of flammable liquids."
G. Signs, signals, barricades	2/4	1926.200. Accident prevention tags; 200(d). "Exit signs, when required, shall be lettered in legible red letters, not less than 6 inches high . . ."
H. Material handling, storage, use	3/3	1926.250. General requirements for storage; 250(b)(1): "Material stored inside buildings under construction shall not be placed within 6 feet of any hoistway or inside floor openings . . ."
J. Welding and cutting	1/5	1926.350. Gas welding and cutting. 350 a(10). "Oxygen cylinders in storage shall be separated from fuel-gas cylinders., a minimum distance of 20 feet (6.1 m) . . ."
L. Scaffolds	2/4	1926.451. General requirements for scaffolds; 452f(1). "Scaffolds shall not be constructed. more than two tiers or 10 feet (3.0 m) in height, whichever is less."
M. Fall protection	2/4	1926.502. Fall protections system criteria and practices; 502j(3). "Toeboards shall be a minimum of 3 1/2 inches (9 cm) in vertical height from their top edge . . ."
N. Hoists, elevators, conveyors	2/7	1926.552. Material hoists, personnel hoists, and elevators; 552c(4): "Hoist doors or gates shall be not less than 6 feet 6 inches high and shall be provided with mechanical locks . . ."
O. Motor vehicles, equipment	5/7	1926.600. Equipment; 600. a(6). "Combustible and flammable materials shall be removed from the immediate area prior to operations."
P. Excavations	2/3	1926.651. Specific requirements; 651(l): "... Guardrails . . . shall be provided where walkways are 6 feet (1.8 m) or more above lower levels."
Q. Concrete and masonry construction	3/6	1926.702. Requirements for equipment and tools; 703(c)(3): "All vertical slip forms shall be provided with scaffolds or work platforms where . . . work or pass."
R. Steel erection	7/12	1926.754. Structural steel assembly; 454.b.(3). "A fully planked or decked floor or nets shall be maintained within two stories or 30 feet (9.1 m), . . ."
T. Demolition	6/11	1926.852. Chutes; 852(e): "Any chute opening, into which workmen dump debris, shall be protected by a substantial guardrail approximately 42 inches above the floor or other surface on which the men stand . . ."
V. Electric power distribution	3/19	1926.958. Materials handling and storage; 958.b(1). "For lines and equipment energized at 50 kilovolts or less, the distance is 3.05 meters (10 feet)."
X. Stairways and ladders	2/5 1R	1926.1053. Ladders; 1053(a)(4)(ii) "The minimum clear distance between side rails for all portable ladders shall be 11 1/2 inches (29 cm)."
AA: Confined spaces	1/13 1R	1926.1203. General requirements; 1203.e(2)(ii): "When entrance covers are removed, the opening must be immediately guarded by a railing, . . ."
CC. Cranes and derricks	10/43	1926.1435. Tower cranes; 1435(b)(6) "... the cranes must be located such that no crane can come in contact with the structure of another crane."

Note: # = number of sections in each subpart that can be represented in BIM over all sections.

^aR: reserved.

typically missing but can be represented in 3D models (e.g., scaffolding, safety nets). Based on the screening results, the research team identified 20 subparts and 91 sections within OSHA Part 1926 that contain narratives that refer to components that can be represented in 3D models. Out of all subparts, the following subparts generally contained regulatory and provisional narratives that could not be tied to potential model elements and were excluded from further analysis: A-general, B-interpretations, I-hand tools, K-electrical, S-underground construction, U-blasting, W-rollover practice structures, Y-commercial diving, Z-toxic substances, and BB-reserved. Examples of the sections that were identified to be further analyzed are provided in Table 1, where at least one narrative in a section had the potential to be checked using BIMs.

In other words, each of these narratives referred to a physical component (or a set of components) in a design model (e.g., trench excavation area), or its attributes (e.g., depth of excavated area) where specific safety equipment/tools/precautions should be provided.

Detailed Analysis of OSHA Safety and Health Regulations for Construction (Part 1926)

The identified 91 sections (e.g., §1926.250 - General requirements for storage, §1926.502 - Fall protection systems criteria and practices, etc.) contained around 275 narratives that tie back to model elements (or potentially to be modeled in BIMs such as a trench area). Narratives typically contained multiple statements that would correspond to multiple rules to be checked.

Development of Rulesets from Narratives

With the safety requirements identified from the previous steps, they were converted into computer-interpretable rulesets in pseudocode form. The safety-requirement narratives are first decomposed into simpler rules to decrease the complexity. Then, these simple rules are converted to implementable and machine-readable rulesets in the form of pseudocodes. For example, narrative 1926.502 (b)(11) states that, “When guardrail systems are used at holes, they shall be erected on all unprotected sides or edges of the hole” which can be converted to a ruleset such as [The component type= “Guard Rail” shall touch the “Opening” on all unprotected sides]. The identified safety requirements from the previous step were converted to computer-processable rulesets. Examples of the narratives stated in OSHA 1926.151 as the safety requirements and the corresponding pseudocodes are shown in Table 2.

As shown in Table 2, each narrative was simplified by decomposing the narrative to simple rulesets that contained logical relationships. In the process of narrative conversion, the capability of the model-checking tool’s built-in rules was also considered so that the rules could be implemented in the selected environment.

Implementation of Defined Rulesets in BIMs

The pseudocodes from the previous section were used to implement them in a model-checking application. The authors evaluated several model-checking tools and decided on using the application that could receive a model in an interoperable format. The application could receive a widely used data format (i.e., IFC) instead of specific BIM authoring tool formats.

Four different configurations of BIMs (Fig. 2) were cleaned to prepare a phased model for construction, where for some of the narratives a working surface/deck had to be defined, confined spaces need to be labeled, and/or holes/openings need to be cut in decks. Different models provided us with base models to modify where specific component types that are needed for specific operations were available (e.g., concrete versus steel, interior versus exterior work, decks with openings). These base models have been modified to represent additional safety related components (e.g., safety nets, guardrails, ladders) as needed.

Rulesets were implemented and successfully checked in the model-checking tool. Fig. 3 shows example violations. A violation of 1926.151(a)(2): “Internal combustion engine powered equipment shall be so located that the exhausts are well away from combustible materials. When the exhausts are piped to outside the building under construction, a **clearance of at least 6 inches** shall be maintained between such **piping** and **combustible material**.”

Table 2. Examples from OSHA 1926 narratives that were converted to rulesets

OSHA section	Fall protection systems criteria and practices	Applicable rules in pseudocode
1926.151 (a)(2)	“Internal combustion engine powered equipment shall be so located that the exhausts are well away from combustible materials. When the exhausts are piped to outside the building under construction, a clearance of at least 6 inches shall be maintained between such piping and combustible material .”	The distance between the componentType= “Combustion Engine Exhaust Pipes” and the componentType = “Combustible Material” shall be > = 6 inches
1926.151 (c)(5)	“Method of piling shall be solid wherever possible and in orderly and regular piles. No combustible material shall be stored outdoors within 10 feet of a building or structure .”	MIN distance between the componentType = “Stored Combustible Material” and any componentType = 10 feet
1926.151 (c)(6)	“Portable fire extinguishing equipment, suitable for the fire hazard involved, shall be provided at convenient, conspicuously accessible locations in the yard area. Portable fire extinguishers , rated not less than 2A , shall be placed so that maximum travel distance to the nearest unit shall not exceed 100 feet .”	The spaceType = “Yard Area” shall contain componentType = “Fire Extinguisher 2A” AND The distance between the componentType = “Fire Extinguisher 2A” and the componentType = “Storage Unit” = <100’
1926.151(d)(7)	“A clearance of 24 inches shall be maintained around the path of travel of fire doors unless a barricade is provided, in which case no clearance is needed. Material shall not be stored within 36 inches of a fire door opening.”	The MIN distance between any componentType = “Stored Material” and the openingType = “Fire Door Opening” = 36”

Note: Bold indicates parameters that are essential in defining a rule. They indicate component types, attributes, or threshold values.

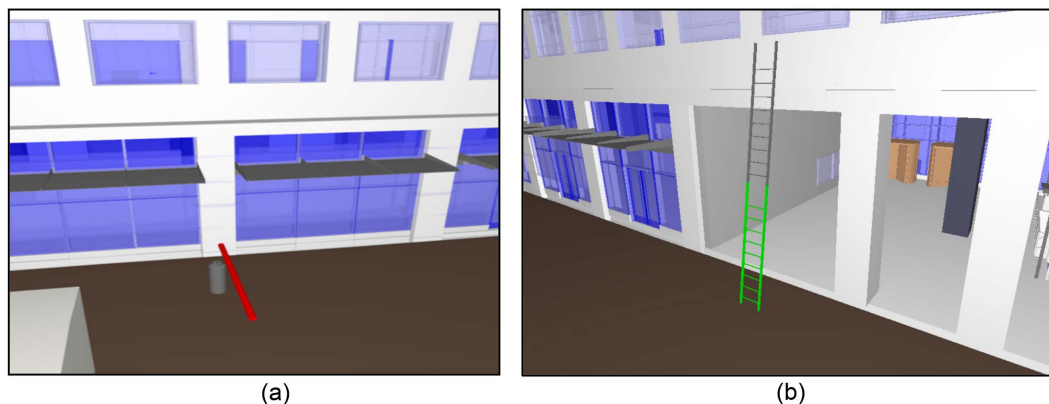


Fig. 3. Example violations. (a) Violation of no combustible material near exhaust pipes; and (b) violation when ladders are fastened together to provide longer sections.

the building under construction, a clearance of at least 6 in shall be maintained between such piping and combustible material” indicated that the equipment was located within 6 inches of the exhaust piping. The figure also shows a violation of 1926.1053(a)(7): “Ladders shall not be tied or fastened together to provide longer sections unless they are specifically designed for such use,” where two ladders were stacked.

Challenges in Achieving Model-Based Safety Checking and Potential Solutions

The results showed that if the rulesets are implemented in the model-checking tools, the tool can successfully check the violations automatically and save tremendous amounts of time and human efforts. However, along the way various types of challenges were identified during the process of the safety requirement identification and ruleset development. These challenges are provided in Table 3 with their definitions.

Identified challenges fall mainly under four categories: challenges that are related to model content, nature of the narratives, model-checking tools, and construction. Under model content, the challenges are mainly due to missing (i.e., components are not in the model), incomplete (i.e., component is there but required attribute is missing), or incorrect representations (i.e., whole-part relationships are not defined for components) in models. Narrative specific challenges are due to (1) having narratives without a clear language that indicates thresholds/boundaries that should be implemented (i.e., vagueness in narratives); (2) having narratives with compounding requirements, where decomposition is needed for proper implementation; or (3) having narratives that are mainly provisional and cannot be linked to model components. Tool related challenges can be faced while implementing model-based safety requirements checking, and these are mainly (1) due to inflexibility of tools in their library of logical operators required for properly implementing the OSHA 1926 narratives; and (2) due to lack of a mapping classification for representing objects that are uncommon in 3D models but needed in narratives through the use of common object types (where this classification will help to map

these uncommon objects always to those common objects) to bring consistency to ruleset implementation. Construction phasing is another challenge, where narratives refer to a specific phase where certain safety requirements be present on the site and the models need to be prepared to enable model-based checking. Fig. 4 provides the main contribution of this work, which is the classification of challenges in the way to model-based safety compliance checking. The details of these challenges and potential viable solutions to eliminate (or have a workaround to these challenges) are provided in the following subsections.

Model Specific Challenges: Missing Components in BIMs and Potential Solutions

Although design models are available to construction teams at first, these models should be converted to construction models, which require the representation of construction equipment (e.g., cranes, hoists), temporary structures (e.g., trenches, scaffoldings, ladders), and safety components (e.g., guardrails, safety nets) that are typically tied to OSHA safety narratives. There are various narratives that refer to components that are typical in construction models, and require additional information to be represented for those components, such as their locations, minimum/maximum distance between them, and their properties that are referred to in the narratives. Table 4 shows the types of additional components that are referred in OSHA narratives and whether these components are typical to have in design BIMs.

As seen in Table 4, the majority of components referred in narratives are atypical in design models and require additional effort by construction teams to represent them. Models need to be transformed from design models to construction models. When 275 narratives that were within the scope of this study were further investigated with respect to the required components, we identified that 90% of components referred in narratives are not in design BIMs and need to be added while safety plans are prepared. Examples are provided in Fig. 5.

Potential solutions: Models need to be transformed from design models to construction models when there is an intention to use the models for model-based safety compliance checking.

Table 3. List of identified challenges and their definitions

Category	Identified challenge	Description
Model	Missing components in BIMs	Indicates that the narratives refer to components that are not part of the design model. e.g., ladders, reshores, safety nets.
	Missing attributes of components	Indicates that the narratives refer to specialized attributes of components that are not part of the design model. e.g., breaking strength of edge ropes in safety nets.
	Missing component decompositions	Indicates that the narratives refer to subparts of components and might not be readily available in models. Components are represented as a single whole object. e.g., rest platform of ladders, bars and gates of hoists.
Nature of narrative	Vagueness in narratives	Indicates that the narratives are not clear in language and can be subjectively interpreted. e.g., pneumatic power tools to be secured by some positive means to keep connected.
	Compound nature of narratives Narratives not linkable to model elements	Indicates that a given narrative in this category requires at least two rules to be adequately represented. Indicates that the narratives cannot be linked to the model as there are no ways such information can be linked to any model component. e.g., recording and reporting of injuries.
Tool	Inflexibility of tools for extending ruleset operators	Indicates that the narratives require complex logical and mathematical operators that a model-checking tool cannot host for representing the ruleset at once.
	Lack of consistent object mapping terminology	Indicates that the narratives refer to an atypical object in BIM authoring tool family of objects. This either necessitates the definition of a new family definition, or using a specified object type in available families to map to. e.g., representation of waste material chutes as a collection of floor and wall objects; guardrails as walls.
Construction	Missing phasing of temporary components in BIM	Indicates that the narratives refer to specific timeframes when temporary structures will be needed. Models need to be prepared to indicate phasing. e.g., defining the working deck for a safety narrative.

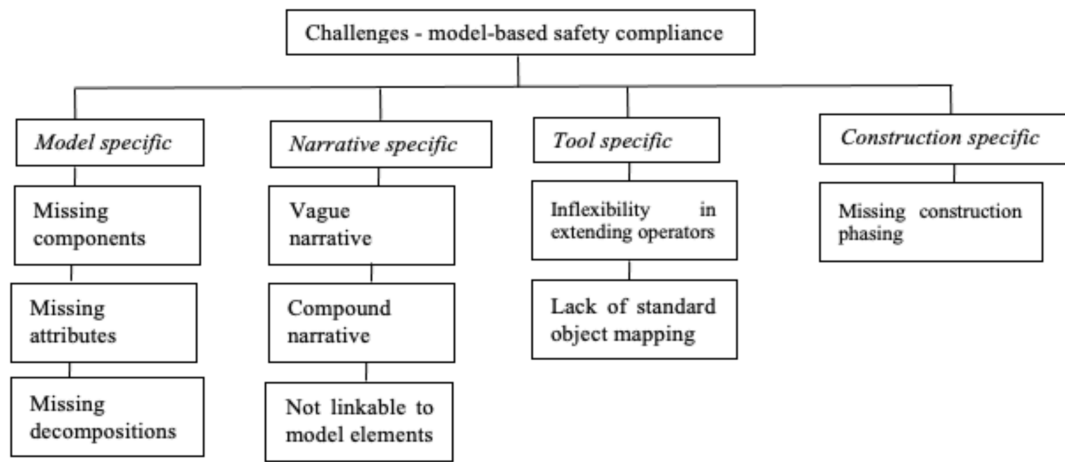


Fig. 4. Classification of challenges in the way to model-based safety compliance checking.

Table 4. Components mentioned in OSHA 1926 subparts and their representations in design models (marked with bold/italic text)

OSHA subparts that can be linked to BIMs	Component types and objects referred in corresponding subparts ^a
C. General safety /health provisions	Exit signs, egress routes
D. Controls	Temporary: Portable containers for water, toilets, temporary sleeping quarters, lavatories, showers, exhaust ventilation systems and related components, room(spray) , lockers, signs
E. Personal protective equipment	Safety nets, ring buoys, lifesaving skiffs
F. Fire protection	Fire extinguishers, water drums, hose lines, combustible material, fire doors , flammable liquids, cabinets (for flammables) , room (storage) , vent piping systems, underground tanks, container valves and accessories, container systems
G. Signs, signals, barricades	Barricades
H. Material handling, storage, and use	Materials stored, disposal systems (chutes) for waste
J. Welding and cutting	Compressed gas cylinders, valves, valve protection caps
L. Scaffolds	Scaffold systems, platforms, scaffold planks, poles, suspension ropes, ladders, stair towers
M. Fall protection	Stair rail systems, hand rail systems, guardrail systems, ramps, vertical lifelines, canopies, safety nets
N. Hoists, elevators, conveyors	Hoists, elevators
O. Motor vehicles, equipment	Vehicles and large construction equipment
P. Excavations	Ramps, ladders, egress routes, sidewalks, walkways
Q. Concrete and masonry	Guardrails, concrete buckets, pumping systems
R. Steel erection	Cranes , wire ropes, hoisting equipment, openings and holes on flat surfaces (decks, roofs), column anchorage , beams and columns , steel joists , working decks (slabs)
T. Demolition	Chutes, ladders, floor openings
V. Electric power distribution	Energized lines and equipment
X. Stairways and ladders	Stairways , ladders
AA: Confined spaces	Space (confined)
CC. Cranes and derricks	Power lines, tower cranes , derricks, side-boom cranes , overhead and gantry cranes

^aBold/italic indicates the components that are typically in a design BIM.

This is a very typical problem when models are used in construction phase, such as for generating 4D models, estimating costs, and coordination and clash detection. In all of these use cases, models will miss some of the critical model elements (e.g., temporary structures needed for scheduling, detailed element compositions for cost estimating, and all model design elements for coordination meetings). Construction teams need to transform original models to construction models to include missing components that are needed in the rulesets. Types of components that are referred to in OSHA 1926 subparts that can be represented in BIMs are provided in Table 4. This list will be handy for researchers and practitioners to populate object libraries in BIM authoring tools to represent these components.

Another way to recognize object types in model-checking tools if these objects are not common is to export them by mapping them to common and reasonable entities in IFC specification.

For example, as shown in Fig. 6, a combustible material is not an instance of an entity that is represented in IFC. It will be wrapped to a proxy entity (e.g., ifcBuildingElementProxy) and will be problematic while implementing rules that ask for combustible material objects. In that case, one can define the entity that this object should be recognized as. In this example, combustible material is mapped to ifcTank in the BIM authoring tool by going through “Export to IFC as” under IFC parameters of object attributes.

In addition to these practical recommendations, there are plenty of existing research studies that provide mechanisms for semantic enrichment in models (i.e., adding semantics related to geometry, nongeometric attributes, and connectivity) to support model use in applications such as in construction management, urban planning, and geoinformatics (e.g., Xue et al. 2021; Dinis et al. 2022; Belsky et al. 2016; Xue et al. 2019; Jiang et al. 2023). As pointed out by Jiang et al. (2023), semantic enrichment studies mainly benefit

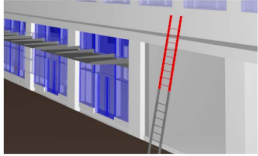
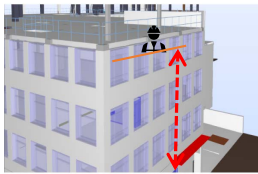
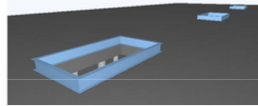
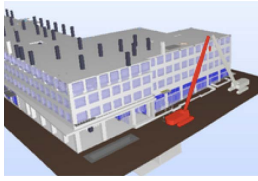
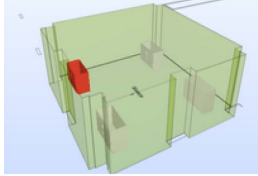
Narrative from OSHA	Implemented ruleset	Required	Visualization of the violation
1926.1053(a)(7): "Ladders shall not be tied or fastened together to provide longer sections unless they are specifically designed for such use."	Distance of object[Type] = <u>Ladder</u> AND object[Type]= <u>Ladder</u> <=2"	Ladders	
1926.502(c)(1): "Safety nets shall be installed as close as practicable under the walking/working surface on which employees are working, but in no case more than 30 feet (9.1 m) below such level."	Distance of object[Type] = <u>safety net</u> AND object[Type]=deck <=30'	Safety nets	
1926.502(b)(1): "Top edge height of top rails, shall be 42 inches plus or minus 3 inches."	Height of objects named/classified as " <u>rails</u> " <= 45" or height >=39"	Guard rails	
1926.1435(b)(6): "On jobsites where more than one crane is installed, the cranes must be located such that no crane can come in contact with the structure of another crane."	Minimum distance between objects named/classified as " <u>cranes</u> " = 2* length of boom	Cranes	
1926.152(b)(3): No more than three storage cabinets (containing flammable liquids) maybe located in a single storage area	Maximum number of components " <u>cabinet</u> " in space named/classified as " <u>storage</u> " = 3	Storage space Cabinets	

Fig. 5. Example narratives requiring additional components in models.

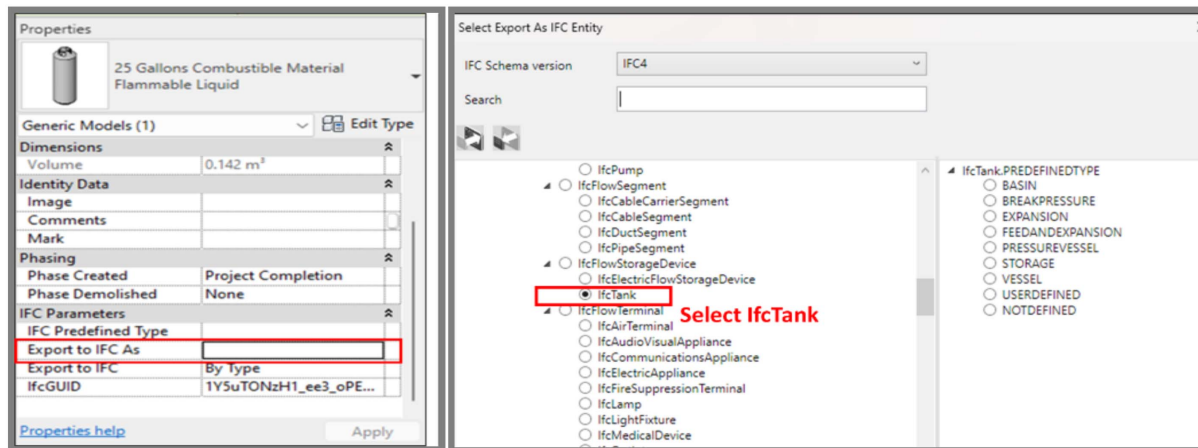


Fig. 6. Mapping an uncommon object type to a specific IFC entity.

from earlier research studies on BIMs with query languages with and without spatial and topological relations (e.g., Mazairac and Beetz 2013; Daum and Borrmann 2014) and rule-checking systems (e.g., Eastman et al. 2009; Pauwels et al. 2011). Scan to BIM to create a geometric model and related technology and solutions

in the literature to convert that geometric model to a semantically rich model serve as potential pathways that can be leveraged here in this context to bring in missing model elements to models for existing and ongoing projects (e.g., Macher et al. 2017). These solutions, however, require that the site be scanned or captured

digitally periodically to reflect progress on the site regarding temporary structures that are referred in safety rules.

Model Specific: Missing Attributes of Components and Potential Solutions

In addition to missing components that are essential for safety compliance checking, there are various specialized attributes needed in models. This was a problem observed both for typical components in models as well as for safety related components that are not typically represented in design models. For example, guardrails, safety nets, and ladders are components that are atypical in design models. Even if they are represented, several of the narratives refer to specialized attributes that should also be represented so the rulesets can be correctly defined. For example, 1926.502(b)(1) asks for representing the value for the ‘top edge height of top rails’ for guardrails, and 1926.1053(a)(3)(i) asks for the ‘spacing between centers of steps’ of portable and fixed ladders to be between 10 and 14 in (see Fig. 7).

Even if the component is represented, if the required attribute in the narrative is not specified with a value, the rule checking will not be complete. In addition, there are cases where typical model elements and their attributes need values, such as a slab to be defined as a ‘working deck’ to specify the location and maximum height from which materials can be dropped. When openings are present

on slabs and used to drop debris without chutes, the edges should be defined for the opening to define the projected edge of the opening on the floor where debris will fall. To solve this problem, modelers should be aware of required attributes as well.

Potential solutions: A direct solution to this problem is to first represent the components in models. The required attributes in narratives will be explicitly needed while implementing the corresponding rulesets. These attributes will be narrative specific and can be available in model-checking tools in multiple ways: (1) attributes that can be explicitly defined in BIM authoring tools, such as *type* of a component (e.g., slab, column); (2) attributes that can be derived from explicit attributes (e.g., deriving area of a slab using its geometric representation); and (3) attributes that are not among commonly shared attributes and require exporting from BIM authoring tools as an IFC property set (e.g., export IFC parameters path in Autodesk Revit). This is mainly for attributes that are not generic but specialized that are represented as property sets, and (4) attributes that are neither generic nor specialized and would require a “tag”/“label” attached to the object so that objects that contain this tag will be considered in the ruleset. Practitioners could check generic and specialized attributes available for a given component type by investigating the type properties in BIM authoring tools they utilize and refer back to related property sets in IFC specification to export them with the needed attributes. In case both paths are void, tagging objects with the needed information

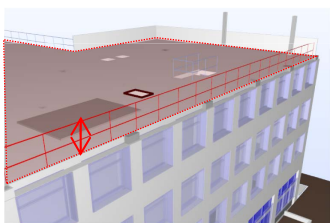
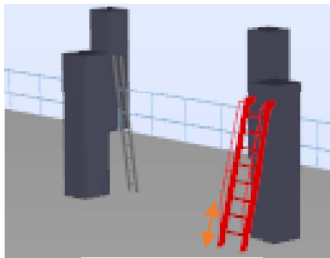
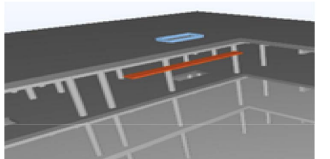
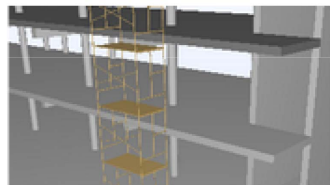
Narrative from OSHA	Ruleset	Attribute	Visualization
1926.502(b)(1): “Top edge height of top rails, or equivalent guardrail system members, shall be 42 inches (1.1 m) plus or minus 3 inches (8 cm) above the walking/working level.”	Height of object[Type]= Top Rail $\leq 45''$ or $\geq 39''$ above object[Type]= Slab	Height of top rail	
1926.1053(a)(3)(i): “Rungs, cleats, and steps of portable ladders shall be spaced not less than 10 inches (25 cm) apart, nor more than 14 inches (36 cm) apart, as measured between center lines.”	For any object[Type] = Ladder Any Distance of object[Type] = Ladder Steps $\leq 14''$ AND $\geq 10''$	Spacing between steps	
1926.105(c)(1): “Safety Nets shall extend at least 8 feet beyond the edge of the work surface”	Width of Object named “safety net” $\geq 8'$	Width of safety nets	
1926.451(b)(2): “Each scaffold platform and walkway shall be at least 18 inches (46 cm) wide.”	Width of any component named “scaffold” $\geq 18''$	Width of scaffold platform	

Fig. 7. Example narratives requiring specialized attributes of components in models.

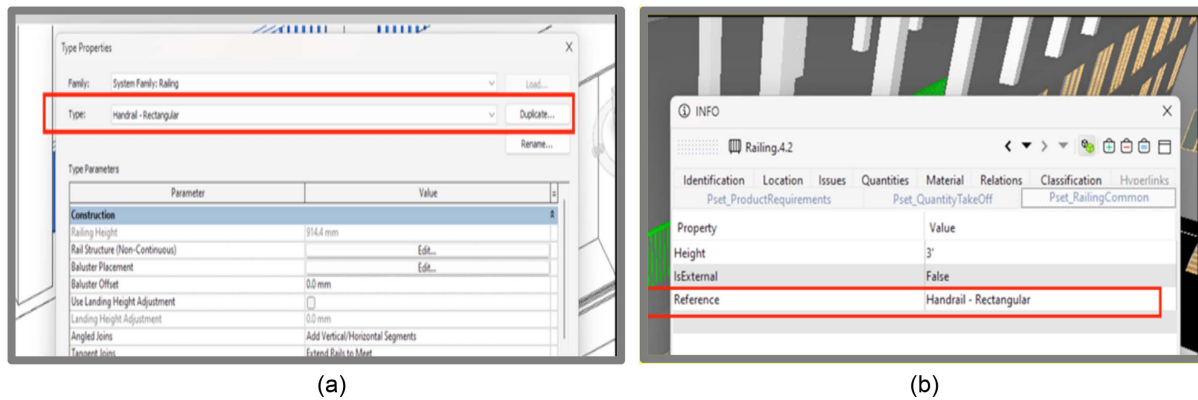


Fig. 8. Solution to add new attributes to components: (a) model element modification in the BIM authoring tool; and (b) model element attributes received in the model-checking tool.

will be a workaround to this problem. An example is provided in Fig. 8 below, where one can export a handrail generic object with a specific subtype needed in a rule (e.g., rectangular). In that case, type attribute can be defined and will be exported with type information available for objects labeled as handrail. The rule can be tied to handrails that have a rectangular type in this case. This solution will also serve as a way around when uncommon objects cannot be mapped to any IFC entities and can be defined with a “type” attribute to tag and separate them from the rest of uncommon objects.

Another route to bring additional parameters to objects from predefined object families is to define new “shared parameters.” This is possible by editing “shared parameters” for a project (e.g., in Revit/Manage/Shared Parameters) and adding new parameters for the project. Then, one can apply suitable shared parameters to predefined families by editing type properties of a selected family and adding required parameters. These new shared parameters can be applied to instances or the type (for any instance of this edited family).

In addition to these practical solutions, existing body of knowledge in the related literature provides solutions to infer information from model elements and add missing semantics to the geometric elements. Several of the semantic enrichment related research studies mentioned in relation to adding missing components will be relevant here to enrich the model elements with the missing attributes data, if implemented. Existing research studies that provide guidance on derivation and/or inference of information from existing data in models (e.g., Bloch and Sacks 2020), representation of concepts that are unique to the application area through ontologies (e.g., heritage artifacts and their features) (e.g., Simeone et al. 2019), and extending the standard representation schemas such as IFCs (e.g., Theiler and Smarsly 2018; Wang et al. 2022) are additional ways to add missing attributes and enrich the semantics of the models used in model-checking practice.

Model Specific: Missing Component Decompositions and Potential Solutions

Model elements require whole/part relationships in various narratives that pertain to safety compliance. If a model contains the component as a single object (e.g., hoist) and the narratives refer to specific subparts (e.g., anchors, tie-ins, wire ropes) of that object for checking its safety compliance, this will be an issue. This limitation poses a constraint on the way to model-based safety compliance checking and requires that components are represented as a

composition of subparts to represent the component of interest. In this case, extra effort is needed during the modeling phase to define parts as separate objects with the object hierarchy maintained.

Several narratives that relate to concrete operations and require the use ladders, fall protection systems, and guardrails refer to subparts of related components. For example, 1926.1053(a)(4)(ii) refers to side rails of the portable ladder so that distance between two side rails can be measured. 1926.1053(b)(1) also indicates that the ladder *side rails* shall extend at least 3 ft above the upper landing surface, which indicates the representation of height information for the *side rails* of ladders. Whenever there are concrete operations on job sites, protruding reinforcing steel poses danger for workers. 1926.701(b) indicates that all protruding steel shall be guarded to protect fall onto or into reinforcing steel. This narrative requires the cast in place concrete components to be represented with separate reinforcement components, which is not the typical practice. Fig. 9 provides similar examples of narratives where subparts of components are referred.

Potential solutions: For those narratives that refer to specific parts of components, representing the component as a single object will not solely help with model-based safety compliance checking. Unless each part of a component referred in narratives is modeled separately, there is no direct solution to this problem. However, potential solutions that generate redundancies but help the rulesets work can be implemented as the following: (1) evaluating whether the attributes referred for subparts (e.g., height of top rail in a ladder) in a narrative can be considered as the attributes of the component (e.g., ladder) itself. This is only possible if objects are geometrically close to bounding boxes (e.g., widths stay the same across the object faces) and not irregular. And (2) evaluating rules for the component itself and for all subparts (e.g., checking existence of toe boards across all platforms of scaffolding instead of checking it for the top one). This will generate a redundant set of issues but will enable the rule to be checked against the related components instead of overlooking them.

Narrative Specific: Vagueness in Narratives and Potential Solutions

Automated compliance checking requires that the language used in the narratives can be directly converted into rules without subjective interpretations and ambiguity. Any wordings used in narratives such as enough, some, and proper would be unclear and open to interpretations, and could result in subjective selections of values to define ‘enough’, ‘some’, or ‘proper’. Such narratives, even if the

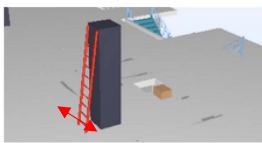
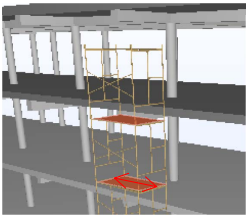
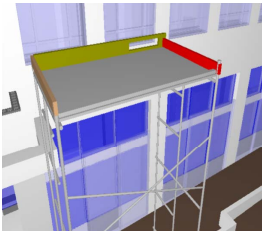
Narrative from OSHA	Ruleset	Subparts	Visualization
1926.1053(a)(4)(ii): "The minimum clear distance between side rails for all portable ladders shall be 11 1/2 inches (29 cm)."	Height of object[Type]= <u>Top Rail</u> ≤ 45" or ≥ 39" above object[Type]= Slab	Ladders and their top rails	
1926.451(h)(2)(ii): "Each scaffold platform and walkway shall be at least 18 inches (46 cm) wide."	Width of object[Type]= <u>walkway/platform</u> OF object[Type]= Scaffold ≥ 18"	Scaffolds and their platforms	
1926.451(b)(2): "A toeboard shall be erected along the edge of platforms more than 10 feet (3.1 m) above lower levels for a distance sufficient to protect employees below, except on float (ship) scaffolds where an edging of 3/4 x 1 1/2 inch (2 x 4 cm) wood or equivalent may be used in lieu of toeboards"	IF height_OF object[Type]=P platform/walkway OF object[TYPE] = Scaffold ≥ 10' All <u>sides</u> OF object[TYPE]= <u>platform</u> HAS object[TYPE]= <u>toe board</u>	Scaffolds, their platforms and toeboards	

Fig. 9. Example narratives requiring subparts of components to be represented in models.

corresponding components, subparts, and attributes are in the model, pose a challenge to accurately convert the narratives to rulesets. The 1926.1435I(5)(i) option indicates that the "trolley rope must be marked (so it can be seen by the operator) at a point that will give the operator sufficient time to stop the trolley prior to the end stops" for operation of derricks. This narrative can be interpreted in many ways to define the location of the marking and the sufficient time for operators. Similarly, 1926.1436(c)(2)(iv) states that for guy derricks "the mast base must permit the mast to rotate freely with allowance for slight tilting of the mast caused by guy slack." In this version, the tilting amount and what is considered as 'slight' is left to the discretion of the interpreter and poses ambiguity. Another example is 1926.754(e)(2)(ii) for hole covers during structural steel construction, where it states that "roof and floor holes and openings shall be decked over. Where large size, configuration or other structural design does not allow openings to be decked over (such as elevator shafts, stair wells, etc.) employees shall be protected in accordance with 1926.760(a)(1)."

Potential solutions: Not many examples are observed in OSHA Part 1926 where a narrative is vague or with missing details about what thresholds are. In these cases, the implementers should set a range of thresholds that are acceptable for the site they are working on to check these narratives. Such narratives will be treated as context dependent, and the thresholds are left to the discretion of the safety managers. In certain cases, the vagueness is eliminated when multiple narratives are combined (subsequent narrative provides more specifics). For example, 152(b)(3) only talks about having maximum cabinets in storage rooms. However, when combined with earlier narratives [e.g., 152(b)], it is clear that this narrative is referring to only cabinets that contain flammable liquids. In these cases, implementers should consider reading related narratives together to see if combined narratives provide the details needed to implement the ruleset properly.

Narrative Specific: Compound Nature of Narratives and Potential Solutions

This challenge indicates that a narrative given under a section contains information for representation of multiple rulesets (i.e., a narrative resulting in at least two or more rule representation). As a result, the number of requirements to be implemented becomes even larger than the OSHA narratives. This indicates that when a model-based process is adopted, the number of rules to check will be larger, however, with an expectation that it will take a shorter time to check their compliance than the current manual practice.

There are several examples observed within OSHA Part 1926. An example is 1926.502(b)(1) for fall protection systems with guardrail systems where "top edge height of top rails, or equivalent guardrail system members, shall be 42 inches (1.1 m) plus or minus 3 inches (8 cm) above the walking/working level. When conditions warrant, the height of the top edge may exceed the 45-inch height, provided the guardrail system meets all other criteria of this paragraph. Note: When employees are using stilts, the top edge height of the top rail, or equivalent member, shall be increased an amount equal to the height of the stilts." Here in this single narrative, one rule has to be defined to check the top edge height of the top rail of a guardrail system (should be within 39–45") and another one for the use of stilts, where the height should be checked with respect to combined height including stilt heights. Regarding safety nets usage for personal protection, 1926.105(d) requires multiple rules to be represented that correspond to this narrative. The narrative indicates that "the mesh size of nets shall not exceed 6 inches by 6 inches. All new nets shall meet accepted performance standards of 17,500 foot-pounds minimum impact resistance as determined and certified by the manufacturers, and shall bear a label of proof test. Edge ropes shall provide a minimum breaking strength of 5,000 pounds." This narrative will be broken into one rule to check

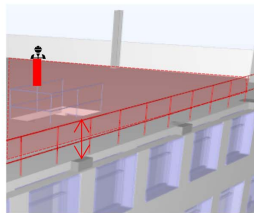
Narrative from OSHA	Implemented ruleset	Visualization of a violation
1926.502(b)(1): "Top edge height of top rails, or equivalent guardrail system members, shall be 42 inches (1.1 m) plus or minus 3 inches (8 cm) above the walking/working level. Note: When employees are using stilts, the top edge height of the top rail, or equivalent member, shall be increased an amount equal to the height of the stilts."	(1) Height of object[Type]= Top Rail ≤ 45 " or ≥ 39 " above object[Type]= Slab AND (2) IF stilt of object[TYPE] = Guardrail available THEN Height of object[Type]= Top Rail $\leq \{45 + \text{height of object[Type]= stilt}\}$ or $\geq \{39 + \text{height of object[Type]= stilt}\}$ above object[Type]= Slab " above object[Type]= Slab	
1926.105(d): The mesh size of nets shall not exceed 6 inches by 6 inches. All new nets shall meet accepted performance standards of 17,500 foot-pounds minimum impact resistance as determined and certified by the manufacturers, and shall bear a label of proof test. Edge ropes shall provide a minimum breaking strength of 5,000 pounds.	(1) Size of object[Type]= mesh of object[Type]= Safety net $\leq 6' \times 6'$ (2) impact resistance of object[Type]=mesh of object[Type]=Safety net $\geq 17,500$ (3) breaking strength of object[Type]=edge rope of object[Type]=Safety net $\geq 5,000$	

Fig. 10. Example narratives requiring multiple rulesets to be defined.

the mesh sizes, one to check the net performance on impact resistance, and one to check the edge ropes' breaking strength. Examples are provided with visualizations in Fig. 10.

Potential solutions: For this particular challenge, the implementers will need to decompose the narrative into implementable rules. Given that none of the information is missing or vague from an implementation perspective, this challenge is only affecting time to implement and get a complete representation of the narrative. There will be one too many mappings on these instances, but the time to check these rules will be automated and will overcome the initial time invested.

Narrative Specific: Not Linkable to Any Components in BIM and Potential Solutions

This challenge is related to narratives that do not refer to any component type (whether within design or construction models) in BIMs. These narratives are typically related to general safety and health provisions (e.g., safety training, recording and reporting injuries, fire protection, first aid, environmental controls). Although there are a handful of those that still can be represented in models (e.g., sanitation containers), several of them are actions that should be taken and are not related to any aspect of model elements, their attributes, or decompositions.

Examples of these include 1926.23, where first aid services and provisions are made available for every employee; 1926.28(a), where employers are required to provide PPE; and 1926.21(b), where employers are held responsible to instruct and train employees for various unsafe conditions, potential hazards, and handling of toxic and flammable materials. Examples of such narratives that have been implemented as comments are provided in Fig. 11.

Potential solutions: This challenge is critical since we want to make sure that all narratives in the 1926 subparts are properly represented in a model-based rule-checking tool. There are two categories of narratives that cannot be linked to model elements. The first category of provisions is not possible to represent in models at all as there are no specific objects that are referred in the narratives. These narratives are general provisions to keep a site safe and healthy. In those cases, these are excluded from the model-checking scope and treated separately. Out of all subparts in 1926, it has been identified that A-general, B-interpretations, I-hand tools, K-electrical, S-underground construction, U-blasting, W-rollover practice structures, Y-commercial diving, Z-toxic substances, and BB-reserved contain narratives that do not refer to specific objects and their implementations on sites, and are to be excluded from the model-checking scope but should be handled as a list of narratives that should be implemented through regular means and methods. The second category of provisions refers to objects (either already a common object represented in a model, or possible to represent in a model) but does not provide specific requirements to be checked against these objects and their proximities to other objects (e.g., who is eligible to operate a power operated tool). These objects can be placed on the model randomly but with a suitable timeframe in the schedule where such tools will be needed (i.e., representation of models for each phase of construction). Given a phase of construction, these objects will be checked, and corresponding notes/comments will be displayed to safety managers as warnings they need to follow up and implement on site. This particular case of provisions will not result in a set of violations observed in the model (since there is no rule specifically implemented due to lack of additional information), but it will result in warning messages displayed for awareness and follow-up actions, when related objects are present in the models.

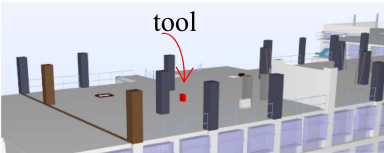
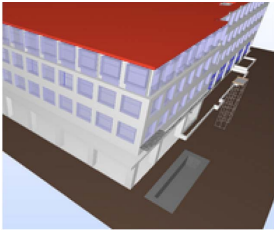
Narrative from OSHA	Implemented comment/note	Visualization of a violation
1926.302(c)(1): for power operated hand tools: “Only employees who have been trained in the operation of the particular tool in use shall be allowed to operate a power-actuated tool.”	A comment is received for every object placed in the model as a power operated tool: “Check if employees are allowed to operate the tool.”	
1926.501(b)5: Each employee on the face of formwork or reinforcing steel shall be protected from falling 6 feet (1.8 m) or more to lower levels by personal fall arrest systems, safety net systems, or positioning device systems.	A comment is received for every object that are steel and concrete and above 6” from Level 0: “Each employee on the face of formwork or reinforcing steel shall be protected from falling 6 feet (1.8 m) or more to lower levels by personal fall arrest systems, safety net systems, or positioning device systems.”	

Fig. 11. Example narratives that cannot be linked to model elements.

Tool Specific: Inflexibility of Tools in Extending Ruleset Operators and Potential Solutions

One of the challenges that pertain to the model-checking tools used would be the extensibility of the library of logical and mathematical operators. Existing off-the-shelf tools might have inflexibility in extending them and might have limited range of logical operators and conditional statements that can be combined together. This challenge was experienced when narratives required multiple conditions to be checked together. For example, 1926.1053(a)(13) is concerned with ladders. It indicates that “the minimum perpendicular clearance between fixed ladder rungs, cleats, and steps, and any obstruction behind the ladder shall be 7 inches (18 cm), except in the case of an elevator pit ladder, for which a minimum perpendicular clearance of 4 1/2 inches (11 cm) is required.” The common method for converting this requirement to a rule would be as shown in Fig. 12 (left) if the tool allowed to combine them in a set of if-then statements and logical operators. However, it was implemented as in Fig. 12 (right). Similarly, OSHA subpart 1926.651(c)(2) deals with the means of egress from trench excavations. It states, “A stairway, ladder, ramp or other safe means of egress shall be located in trench excavations that are 4 feet (1.22 m) or more in depth so as to require no more than 25 feet (7.62 m) of lateral travel for employees.” State of the art off-the-shelf

model-checking tools did not provide mechanisms to combine these conditions as if-then statements and resulted in decomposition of this narrative into multiple rules.

Potential solutions: In addition to creating more rules, this limitation results in more false negatives when the BIM model was checked for the safety requirements. The model-based safety process would necessitate tools that are more flexible to define new rulesets or in-house prototypes need to be developed that parse information exchanged for a BIM and check the necessary components.

Various research studies aimed at addressing this problem and proposed approaches beyond checking OSHA requirements and focused on requirements such as accessibility, building design code regulations, and circulation/egress. These include structured vocabularies, ontologies, logic operators, and semantic similarity to decompose narratives and normative text into a set of structured rules and understand the text (e.g., Hjelseth and Nispet 2011; Mendonça et al. 2020; Zhou and El-Gohary 2021). For instance, RASE methodology (Hjelseth and Nispet 2011; Mendonça et al. 2020) uses four operators: requirement (R), indicating the basis for a defined rule with thresholds, expected outcomes, etc., typically identified with *must* or *shall* wordings in a given narrative; applicabilities (A), indicating the component/object that this rule

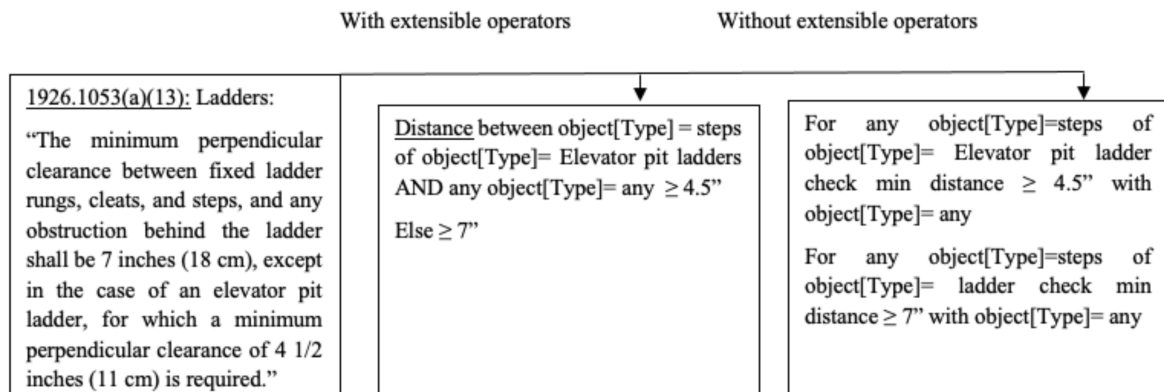


Fig. 12. Inflexible tools for extending library of operators.

applies to; selection (S), indicating whether there is any subset referred in the narrative for the component/object of interest; and exceptions (E), indicating conditions where the requirement does not apply or is excluded from the scope, to transform the narrative in documents into well-defined rules that can be implemented in model verification software based on BIM/IFC. Another way is to use the Linked Data representation (Hjelseth 2016; Kovacs and Micsik 2021), which can provide data interoperability for various software tools, with the use of core vocabularies to describe Persons, Businesses, Locations, Criteria, etc. Linked Data can leverage the resource description framework (RDF) developed for data interchange on the Web and identify a source by a URI to decompose a narrative into rulesets. Specific to OSHA safety requirements, Zhang et al. (2013) introduces a domain-specific language as a language-driven approach to the rule checking and design analysis along with the rule-checking software as a potential remedy for this challenge. Model-checking tools always improve with newer versions, and logical operator libraries will be expected to evolve.

Tool Specific: Lack of a Consistent Object Mapping Terminology and Potential Solutions

This bottleneck is related to components that are not typically in the object libraries of BIM authoring tools or data exchange file formats (e.g., in IFC specification). When components referred in the safety narratives are not in modeling tool libraries (e.g., safety nets, toe boards, barricades, chutes), it is possible that modelers use existing object types in lieu of the required component types (e.g., using wall types to represent barricades, using pipe types for chutes). Two problems arise in this practice: first, it results in upbringing the misrepresented components (e.g., walls represented as barricades) when a ruleset refers to the real component type (e.g., walls) and increasing the number of violations (when in fact those are not real violations). Second, it results in missing those violations when checking of rulesets that refer to components that are misrepresented (e.g., rulesets that refer to barricades and barricades represented with another object type and not tagged to distinguish them). Since no such components are present in the

model accurately (e.g., barricades represented as barricades), rules will not find the applicable component type, although they are in the model with incorrect component types (e.g., walls representing barricades). Both cases are problematic for complete safety compliance checking.

Potential solutions: As a possible solution, it is essential that the component types referred in machine readable rules correctly map to the component types modeled in a BIM authoring tool. For example, if there is no family in the BIM authoring tool to define a “safety net” or the data exchange format (e.g., IFC) does not define that component, then modelers can use different types of components (e.g., a mesh) to represent a safety net. One challenge is that these mappings are observed to be project specific or company specific and not standardized across the architectural, engineering, and construction (AEC) industry. A standard mapping terminology is needed so that when replacements are possible, the rules can check those components that are not of the correct type. The result of this will increase the set of violations, but at least will guarantee that no rule is left out without checking them with respect to model components. These examples are suggesting potential representations in lieu of the correct component types; however, these are not standard in the practice, and there is still a need to define a correct mapping terminology to eliminate omission of rules in the safety compliance checking process. As shown previously, one can export these uncommon objects with tagging new attributes (e.g., type) and/or map it to a suitable IFC entity to narrow down the objects to be referred to from the model in a specific rule.

A consistent mapping terminology will minimize mismatches when running the rules against applicable components. Table 5 shows potential mappings from the OSHA narratives to the standard component types when the narratives contain a specialized component type. An industry-wide effort is needed to either extend the BIM authoring tool object libraries to incorporate these model components, or related data standards (e.g., IFCs) to enable mapping of the data to standard object representations, or define a standard mapping terminology that all practitioners use to replace missing component types consistently.

Another solution to this problem is tagging objects before exporting from BIM authoring tools. One can always add additional

Table 5. Examples for potential mappings between the required and standard component types based on component geometric similarities

Component types referred in narratives	Standard component types selected in lieu of the component in narratives	Implementations to recognize the components
Guardrails	Railings	Export to IFC as: ifcRailing Duplicate type name: “Guardrail”
Safety nets	Ceiling grid, open pipe grid ceilings	Export to IFC as: ifcCovering Duplicate type name: “Safety net”
Containers for disposal materials	Furniture objects	Export to IFC as: ifcFurniture Duplicate type name: “Container”
Coverage of holes and openings	Roof, slab	Export to IFC as: ifcSlab Duplicate type name: “Coverage”
Reshores	Wide flange steel columns, steel HSS tubes	Export to IFC as: ifcColumn Duplicate type name: “Reshore”
Exhaust piping	Duct	Export to IFC as: ifcDuctSegment Duplicate type name: “Exhaust piping”
Portable/fixed ladders	Ladders	Export to IFC as: ifcDiscreteAccessory, ifcStairFlight Duplicate type name: “Ladder”

properties to instances and refer to those properties in the model-checking tools in ruleset implementation so that rules refer to only those instances of a common object (that has been used to represent an uncommon object) to be referred during the rule-checking process.

In addition to the above given practical solutions, there is a plethora of existing literature on object classification in relation to BIMs using geometric information. These studies use computational techniques such as deep learning and object classification to improve classification performance to enable interoperability between applications that reason with BIM data (e.g., Luo et al. 2023; Wu and Zhang 2019). Such approaches, when implemented, can provide alternative ways to label the object with the correct object labels.

Construction Specific: Lack of Phasing Information in Models and Potential Solutions

Usage of temporary structures is subject to the phases that they are used in. Hence, to properly check the phase, some rules should be checked, and models should be prepared for each phase. In addition to this, some of the narratives refer to measurements taken with respect to a working surface, which will be dynamic given the construction sequence. When models are imported into model-checking platforms, the files contain model elements that are defined for the 'current' phase, indicating that each phase and scope is checked separately, and the model is not checked as a whole against the rulesets. For example, in OSHA subpart 1926.105(c)(1), "Nets shall extend 8 feet beyond the edge of the work surface where employees are exposed and shall be installed as close under the work surface as practical but in no case more than 25 feet below such work surface." The working surface and the scope of elements that are expected to be constructed in that phase should be clearly marked so that the current working planes can be identified. Scaffolding, nets, chutes, or excavation related safety measures should be checked when these components are expected first at sites, which again requires phasing/sequencing information represented as part of the models.

Potential solutions: A clear solution to this problem is to bring phasing and version control to models developed for site safety. Phasing will help safety managers and inspectors to see where the safety requirements and associated objects will be at each phase of construction. In model authoring tools, it is possible to define a phase for model elements or a version of a model. This is one way of filtering models based on phases (e.g., site preparation, excavation, superstructure below 50', above 50', etc.) defined for a project. Nowadays, city agencies also ask for general contractors to submit their safety plans using BIMs and provide guidelines to facilitate these submissions (e.g., New York City BIM site safety guidelines). Practitioners can benefit from these guidelines to provide models with correct phasing and model components and their required parameters/attributes.

To summarize, in order to achieve a comprehensive model-based safety checking, improvements in all categories of challenges presented (i.e., model specific, construction specific, tool specific, narrative specific) are essential. The narratives could be revised to contain more specific measurements and could be simplified (or further separated) to convert them to implementable rules. On the other hand, BIM editing tools could be improved to include object families with recognizable attributes for components specific to the safety domain. Model-checking tools could be enhanced to include the necessary tools to easily recognize the modeled objects and related attributes and to provide flexibility to extend built-in rules to check the required safety measures.

Conclusion

This research done with the guidance of an insurance company, various construction teams, and detailed analysis of narratives in OSHA Part 1926 resulted in a categorized list of challenges in the way toward fully automated safety compliance checking. The aim is to enable proactive checking of the applicable rules when the next phase of the construction is about to start so any safety related concerns could be flagged as violations for inspectors ahead of construction. The research team analyzed the federal safety requirements (OSHA Part1926) within 29 subparts with respect to their potential for automated compliance checking. The analyzed narratives from OSHA 1926 were then converted to software implementable rulesets, and the rules were checked against the model using a BIM validation tool. The research team identified four major groups of challenges on the way to automated safety checking: model, narrative, tool, and construction specific. Model specific challenges are related to what BIMs provide as data and show that models lack component types that are typical in the safety domain but not part of standard object libraries, lack specialized attributes that are referred in narratives, or lack whole/part representations that are quite referred in OSHA narratives but not practically represented in models. Narrative specific challenges are observed with respect to the vagueness in the language to accurately convert them to rulesets, or the compound nature of narratives that necessitate multiple rulesets to be defined per narrative. Model-checking tools also pose additional constraints on the way to automated safety compliance with their limitations on logical operator selections and modifications, as well as ways to maintain a standard object mapping terminology that can be utilized when component types referred in narratives are not in model authoring tools and are replaced with standard component types. Lastly, construction phasing information is needed for grouping components that will be constructed in the next phase so that related rules can be checked for those components only.

This work provides a taxonomy of challenges in the way to fully automated safety compliance checking using models based on detailed investigation and implementation of OSHA Part1926 narratives under 20 subparts that can be represented in BIMs. This work also provides practitioners and researchers with potential solutions to these challenges so model-based checking can be fully implemented. The outcome is the first one that systematically provides a classification of these challenges. Findings provide a foundation for researchers to provide solutions to eliminate such challenges. Several practitioner groups can benefit from the outcomes of this work. Construction teams would benefit from the outcomes by understanding the challenges and proactively devising alternative paths to solve them if they are adopting model-based compliance checking in their practice. Model-checking and BIM authoring tool vendors can improve their categories of object libraries, object decomposition representations, and representation of specialized attributes, and extend the ruleset representation operators and mathematical formulations. Regulators can benefit from the outcomes to bring more clarity to the language used for defining missing information so that narratives can be objectively converted to rulesets.

Data Availability Statement

All data, models, and code generated or used during the study appear in the published article.

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References

- Behm, M. 2005. "Linking construction fatalities to the design for construction safety concept." *Saf. Sci.* 43 (8): 589–611. <https://doi.org/10.1016/j.ssci.2005.04.002>.
- Belsky, M., R. Sacks, and I. Brilakis. 2016. "Semantic enrichment for BIM." *Comput.-Aided Civ. Infrastruct. Eng.* 31 (4): 261–274. <https://doi.org/10.1111/mice.12128>.
- Benjaoran, V., and S. Bhokha. 2010. "An integrated safety management with construction management using 4D CAD model." *Saf. Sci.* 48 (3): 395–403. <https://doi.org/10.1016/j.ssci.2009.09.009>.
- Bloch, T., and R. Sacks. 2020. "Clustering information types for semantic enrichment of building information models to support automated code compliance checking." *J. Comput. Civ. Eng.* 34 (6): 04020040. [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000922](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000922).
- BLS (Bureau of Labor Statistics). 2017. "Census of fatal occupational injuries summary, 2017." Accessed February 15, 2023. <http://www.bls.gov/news.release/cfoi.nr0.htm>.
- Daum, S., and A. Borrmann. 2014. "Processing of topological BIM queries using boundary representation based methods." *Adv. Eng. Inf.* 28 (4): 272–286. <https://doi.org/10.1016/j.aei.2014.06.001>.
- Dinis, F. M., J. P. Martins, A. S. Guimarães, and B. Rangel. 2022. "BIM and semantic enrichment methods and applications: A review of recent developments." *Arch. Comput. Methods Eng.* 29 (2): 879–895. <https://doi.org/10.1007/s11831-021-09595-6>.
- Eastman, C. 2009. "Automated assessment of early concept designs." *Archit. Des.* 79 (2): 52–57. <https://doi.org/10.1002/ad.851>.
- Eastman, C., J. Lee, Y. Jeong, and J. Lee. 2009. "Automatic rule-based checking of building designs." *Autom. Constr.* 18 (8): 1011–1033. <https://doi.org/10.1016/j.autcon.2009.07.002>.
- Egan, J. 1998. *Rethinking construction: Report of the construction task force*. London: Dept. of Trade and Industry.
- Hinze, J., and F. Wiegand. 1992. "Role of designers in construction worker safety." *J. Constr. Eng. Manage.* 118 (4): 677–684. [https://doi.org/10.1061/\(ASCE\)0733-9364\(1992\)118:4\(677\)](https://doi.org/10.1061/(ASCE)0733-9364(1992)118:4(677)).
- Hjelseth, E. 2016. "Classification of BIM-based model checking concepts." *J. Inf. Technol. Constr.* 21 (Nov): 354–369.
- Hjelseth, E., and N. Nisbet. 2011. "Capturing normative constraints by use of the semantic mark-up rase methodologies." Accessed October 28, 2022. <https://itc.scix.net/pdfs/w78-2011-Paper-45.pdf>.
- Irizary, J., K. L. Simonsen, and D. M. Abraham. 2005. "Effect of safety and environmental variables on task durations in steel erection." *J. Constr. Eng. Manage.* 131 (12): 1310–1319. [https://doi.org/10.1061/\(ASCE\)0733-9364\(2005\)131:12\(1310\)](https://doi.org/10.1061/(ASCE)0733-9364(2005)131:12(1310)).
- Jiang, S., X. Feng, B. Zhang, and J. Shi. 2023. "Semantic enrichment for BIM: Enabling technologies and applications." *Adv. Eng. Inf.* 56 (Apr): 101961. <https://doi.org/10.1016/j.aei.2023.101961>.
- Kim, K., Y. Cho, and S. Zhang. 2016. "Integrating work sequences and temporary structures into safety planning: Automated scaffolding-related safety hazard identification and prevention in BIM." *Autom. Constr.* 70 (Oct): 128–142. <https://doi.org/10.1016/j.autcon.2016.06.012>.
- Kovacs, A. T., and A. Micsik. 2021. "BIM quality control based on requirement linked data." *Int. J. Archit. Comput.* 19 (3): 431–448. <https://doi.org/10.1177/14780771211012175>.
- Ku, K., and T. Mills. 2010. "Research needs for building information modeling for construction safety." In *Proc., Associated Schools of Construction 46th Annual Int. Conf.* Fort Collins, CO: Associated Schools of Construction.
- Li, X., W. Yi, H. Chi, X. Wang, and A. Chan. 2018. "A critical review of virtual and augmented reality (VR/AR) applications in construction safety." *Autom. Constr.* 86 (Feb): 150–162. <https://doi.org/10.1016/j.autcon.2017.11.003>.
- Luo, H., G. Gao, H. Huang, Z. Ke, C. Peng, and M. Gu. 2023. "A geometric-relational deep learning framework for BIM object classification." In *Proc., European Conf. on Computer Vision*. Cham, Switzerland: Springer. https://doi.org/10.1007/978-3-031-25082-8_23.
- Macher, H., T. Landes, and P. Grussenmeyer. 2017. "From point clouds to building information models: 3D semi-automatic reconstruction of indoors of existing buildings." *Appl. Sci.* 7 (10): 1030. <https://doi.org/10.3390/app7101030>.
- Mazairac, W., and J. Beetz. 2013. "BIM-L—An open query language for building information models." *Adv. Eng. Inf.* 27 (4): 444–456. <https://doi.org/10.1016/j.aei.2013.06.001>.
- Mendonça, E., L. Manzione, and E. Hjelseth. 2020. "Converting Brazilian accessibility standard for BIM-based code checking using RASE and SMC." In *Proc., 37th CIB W78 Information Technology for Construction Conf. (CIB W78)*, 291–307. Toronto: International Council for Research and Innovation in Building and Construction.
- NYC DoB (New York City Department of Buildings). 2014. "Building code—Safeguards during construction or demolition (Chapter 33)." Accessed March 10, 2023. <http://www1.nyc.gov/site/buildings/codes/2014-construction-codes.page#bldgs>.
- OSHA (Occupational Safety and Health Administration). 1926. "OSHA construction regulations." Accessed February 26, 2023. https://www.osha.gov/pls/oshaweb/owastand.display_standard_group?p_toc_level=1&p_part_number=1926.
- Pauwels, P., D. Van Deursen, R. Verstraeten, J. De Roo, R. De Meyer, R. Van De Walle, and J. Van Campenhout. 2011. "A semantic rule checking environment for building performance checking." *Autom. Constr.* 20 (5): 506–518. <https://doi.org/10.1016/j.autcon.2010.11.017>.
- Simeone, D., S. Cursi, and M. Acierno. 2019. "BIM semantic-enrichment for built heritage representation." *Autom. Constr.* 97 (Jan): 122–137. <https://doi.org/10.1016/j.autcon.2018.11.004>.
- Solibri. 2021. "Model checking solution." Accessed April 14, 2021. <http://solibri.com>.
- Staub-French, S., and A. Khanzode. 2007. "3D and 4D modeling for design and construction coordination: Issues and lessons learned." *J. Inf. Technol. Constr.* 12 (26): 381–407.
- Sulankivi, K., T. Mäkelä, and M. Kiviniemi. 2009. "BIM-based site layout and safety planning." In *Proc., 1st Int. Conf. on Improving Construction and Use through Integrated Design Solutions*, 125–140. Espoo, Finland: Technical Research Centre of Finland.
- Theiler, M., and A. Smarsly. 2018. "IFC Monitor—An IFC schema extension for modeling structural health monitoring systems." *Adv. Eng. Inf.* 37 (Aug): 54–65. <https://doi.org/10.1016/j.aei.2018.04.011>.
- Vries, B., S. Verhagen, and A. J. Jessurun. 2004. "Building management simulation centre." *Autom. Constr.* 13 (5): 679–687. <https://doi.org/10.1016/j.autcon.2004.03.003>.
- Wang, T., V. Gan, D. Hu, and H. Liu. 2022. "Digital twin-enabled built environment sensing and monitoring through semantic enrichment of BIM with SensorML." *Autom. Constr.* 144 (Dec): 104625. <https://doi.org/10.1016/j.autcon.2022.104625>.
- Wu, J., and J. Zhang. 2019. "New automated BIM object classification method to support BIM interoperability." *J. Comput. Civ. Eng.* 33 (5): 04019033. [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000858](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000858).
- Xue, F., K. Chen, W. Lu, and A. Zetkalic. 2019. "From semantic segmentation to semantic registration: Derivative-free optimization-based approach for automatic generation of semantically rich as-built building information models from 3D point clouds." *J. Comput. Civil Eng.* 33 (4): 04019024. [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000839](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000839).
- Xue, F., L. Wu, and W. Lu. 2021. "Semantic enrichment of building and city information models: A ten-year review." *Adv. Eng. Inf.* 47 (Jan): 101245. <https://doi.org/10.1016/j.aei.2020.101245>.

- Zhang, J., and Z. Hu. 2011. "BIM- and 4D-based integrated solution of analysis and management for conflicts and structural safety problems during construction: 1. Principles and methodologies." *Autom. Constr.* 20 (2): 155–166. <https://doi.org/10.1016/j.autcon.2010.09.013>.
- Zhang, S., K. Sulankivi, M. Kiviniemi, I. Romo, E. Eastman, and J. Teizer. 2015. "BIM-based fall hazard identification and prevention in construction safety planning." *Saf. Sci.* 72 (Feb): 31–45. <https://doi.org/10.1016/j.ssci.2014.08.001>.
- Zhang, S., J. Teizer, J. Lee, C. Eastman, and M. Venugopal. 2013. "Building information modeling (BIM) and safety: Automatic safety checking of construction models and schedules." *Autom. Constr.* 29 (Jan): 183–195. <https://doi.org/10.1016/j.autcon.2012.05.006>.
- Zhou, P., and N. El-Gohary. 2021. "Semantic information alignment of BIMs to computer-interpretable regulations using ontologies and deep learning." *Adv. Eng. Inf.* 48 (Apr): 101239. <https://doi.org/10.1016/j.aei.2020.101239>.